

PROJECT SUMMARY

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Executive Project Summary: 3-Story Steel Frame

Prepared by STRUCTMIND AI

Project Description

This project entails the structural steel detailing, fabrication, and erection of a three-story commercial building. The primary lateral force-resisting system is comprised of steel moment frames, designed to provide an open and flexible interior floor plan. Totalling approximately 180 tons of structural steel, the design, detailing, and fabrication will adhere strictly to the AISC 360-22 Specification and associated standards. The scope covers all primary and secondary steel members required to complete the structural skeleton, preparing it for subsequent trades.

Key Quantities

- Total Tonnage: 180 tons
- Total Pieces (approx.): 2,200 (main members & detail material)
- Field Bolted Connections (approx.): 850 (includes moment, shear, and splice connections)

Top 5 Risks

1. Connection Complexity: The fabrication and inspection of AISC 360-22 compliant moment connections require stringent QA/QC. Any deviation in weld quality or geometry could lead to costly rework or field modifications.
2. Mill Lead Times: Volatility in the steel supply chain may cause delays in material procurement, directly impacting the fabrication schedule and project start date.
3. Erection Sequencing: Moment frames require careful sequencing and temporary bracing for stability during erection. Failure to follow the prescribed plan could compromise safety and structural integrity.
4. Labor Shortage: A lack of certified welders or experienced ironworkers for field erection can create significant schedule delays and increase labor costs.
5. Design Coordination: Incomplete coordination between structural drawings and MEP (Mechanical, Electrical, Plumbing) requirements can lead to conflicts, requiring field modifications to primary steel members.

Top 5 Opportunities

1. **Schedule Acceleration:** The prefabricated nature of a steel frame allows for rapid erection, potentially shortening the overall project timeline and allowing earlier access for interior trades.
2. **Architectural Freedom:** The moment frame system eliminates the need for obstructive diagonal bracing, maximizing usable floor space and providing significant architectural flexibility.
3. **Fabrication Optimization:** Implementing automated or semi-automated welding and cutting processes in the fabrication shop can enhance connection quality, reduce shop hours, and lower costs.
4. **Value Engineering:** Exploring bolted-end-plate or other proprietary moment connection types could reduce complex field welding, accelerating erection and reducing inspection requirements.
5. **Sustainability:** The high recycled content of structural steel and its 100% recyclability at end-of-life can contribute significantly to project sustainability targets, such as LEED certification.

Recommended Schedule Horizon

A total duration of 16-20 weeks is recommended from the release of approved-for-construction drawings to the completion of steel erection.

- Detailing & Approval: 4-6 weeks
- Procurement & Fabrication: 8-12 weeks
- Erection: 2-3 weeks